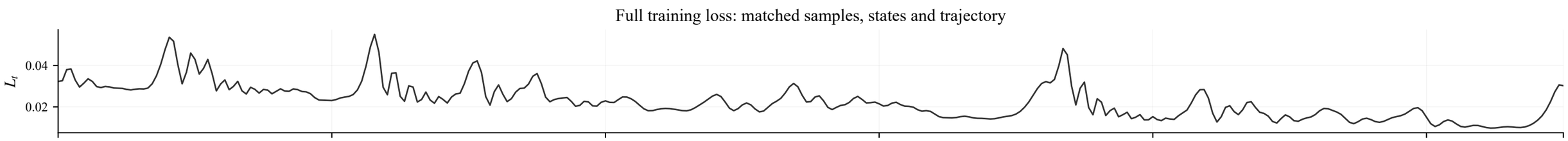
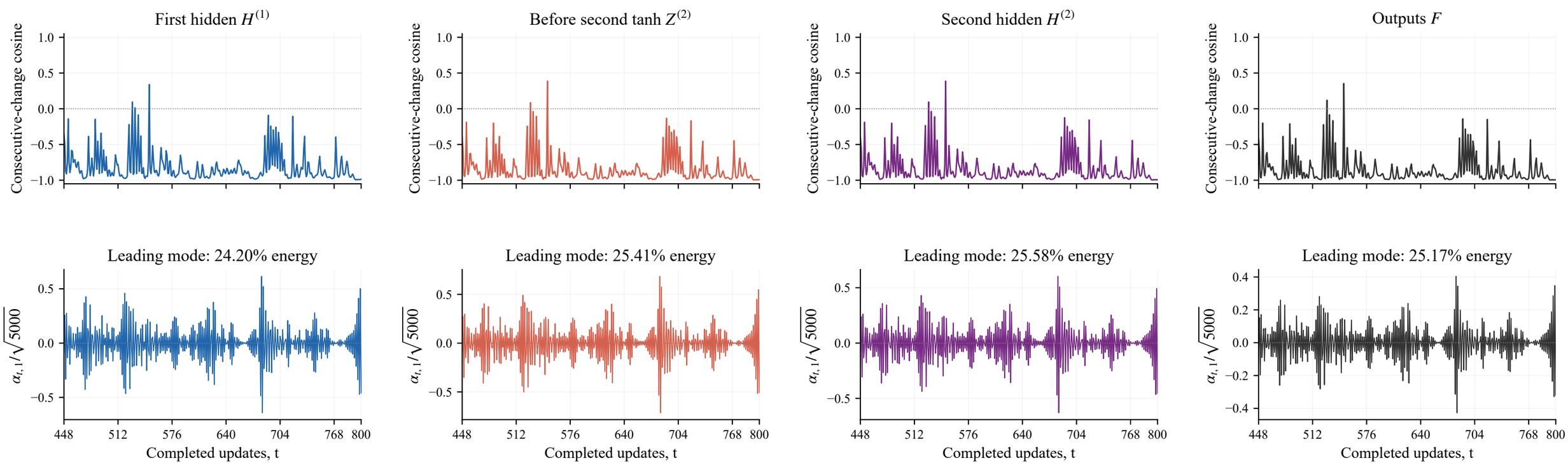
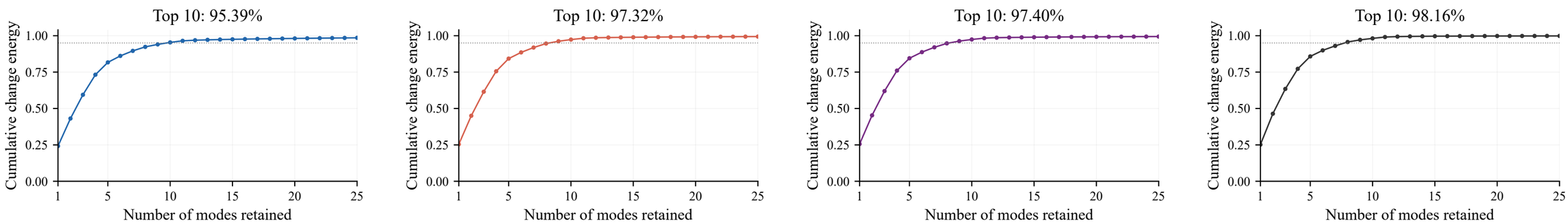
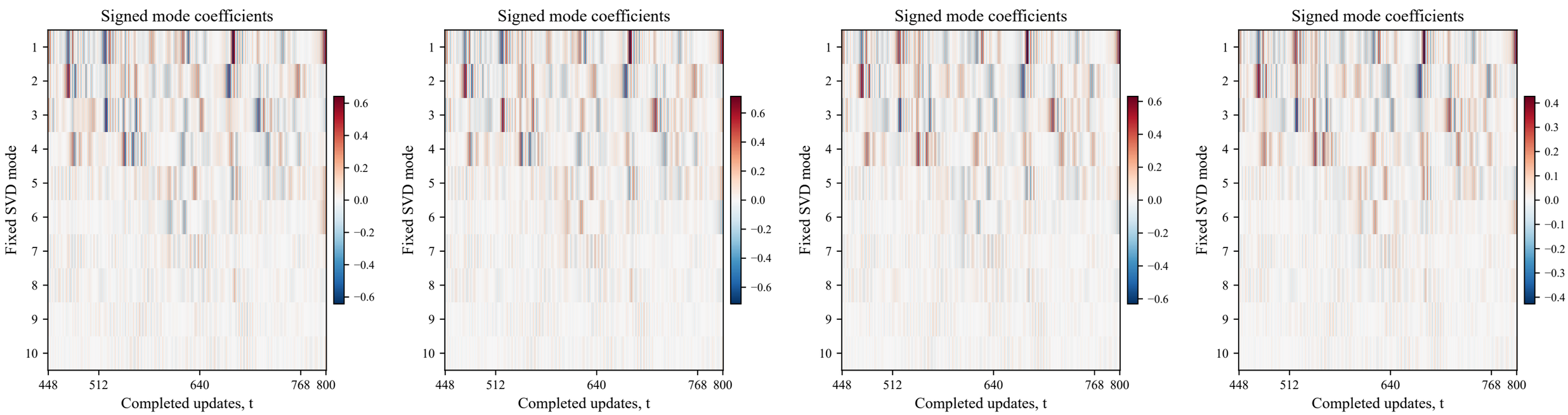
CIFAR-10 MLP: tracing output oscillation backward through both hidden layers

$H^{(1)}$ (5,000 samples x 200 units) ⟶ linear $Z^{(2)}$ (5,000 x 200) ⟶ tanh $H^{(2)}$ (5,000 x 200) ⟶ output F (5,000 x 10)









Each fitted row is $X_{t+1} - X_t$; each cosine compares outgoing and incoming changes around t. All examples are retained.

Fixed uncentered SVDs are fitted independently. Equal widths do not make hidden columns correspond between layers.

Signs are fixed within each basis; mode numbers and coefficient signs across spaces do not imply matched directions.